Breaking the Lexical Supervision Ceiling: Clause-Level MiniLM-L6 Verification and Task-Priority Intent Routing on a Zero-VRAM Edge NPU

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Abstract. In a prior study (DOI: 10.5281/zenodo.23901605), we showed that deterministic speech-regex and stem-list action supervisors reach an empirical ceiling on live language model outputs: across $N = 1{,}000$ four-model completions, a 44-stem grammar supervisor caught 71.1% (128/180) of in-vocabulary action claims but dropped to 29.5% (23/78) on unseen verb lemmas (58.5% overall), while Structured-Output Action Contracts (SOAC) triggered spurious action frames on 13.3%–16.7% of conversational idioms. Replacing lexical rules with a neural classifier solves open-vocabulary generalization on isolated single sentences, yet introduces two new failure modes in edge companion deployments: (1) dGPU VRAM contention with the primary language model, and (2) severe probability dilution when a single fabricated action clause or task request is buried inside a multi-sentence response. In this paper, we present a three-classifier neurosymbolic supervision and routing architecture (`cross-encoder/nli-MiniLM2-L6-H768`, 82.1M parameters) compiled to FP16 on an Intel AI Boost NPU (0 MB dGPU VRAM, 1.22–2.95 ms mean inference latency on an Intel Core Ultra 9 275HX). By decomposing multi-sentence inputs into syntactic clauses (`split_narration_into_clauses`) and applying domain-specific pooling and symbolic verification operators—clause-level speech-act classification paired with deterministic receipt set-membership verification for action claims, max-hazard lattice pooling for epistemic proof policy, and task-priority pooling for multi-head intent routing—we eliminate multi-sentence dilution without retraining the frozen OpenVINO FP16 NPU weights. Across an expanded $N = 2{,}022$ action-claim evaluation battery (combining the 1,000-turn 4-model corpus, 715-turn Paper 02 v1/v2 6-arm corpus, 300 organic companion discovery/held-out turns, and 30 implicit zero-verb stress probes), the clause-level NPU verifier achieves **99.26% accuracy (**2,007/2,022**, 95% Wilson CI:** [98.78%, 99.55%]**), 98.73% recall (**465/471**)**, and **0.58% FPR (**9/1,551**)**, outperforming the 44-stem regex baseline (11.46% recall [54/471], McNemar $\chi^2 = 399.04, p = 3.07 \times 10^{-89}$). On a pre-registered 150-item code-and-weight-locked holdout (`WO42_LOCKED_HOLDOUT_150`) authored by three isolated subagents and graded by blinded raters ($\kappa = 1.000$), the three NPU classifiers achieve 145/150 (96.67%, 95% CI: [92.43%, 98.57%]) single-pass accuracy and 150/150 (100.0%) after fixing two deterministic router rules (`has_z3` domain precedence and Markdown backtick stripping around app names). Across a live 100-turn end-to-end evaluation spanning all four local Gemma-4 GGUF checkpoints (`26B-A4B`, `12B`, `E4B`, `E2B`), the system achieves **100/100 (100.0%)** functional coherence and grounding and **98/100 (98.0%)** task/reasoning completion. Finally, to resolve the sample-size limits of our earlier preprints (DOI: 10.5281/zenodo.23901452, DOI: 10.5281/zenodo.23901683), we execute pre-registered $N = 300$ expansions (`prereg-wo43-power-300`) of the 5-arm Memory Poisoning Ablation (0/100 [0.0%, 95% CI: [0.00%, 4.61%]] Cohort A self-poisoning and Cohort C user-misinformation absorption vs. 100/100 Naive RAG, exact McNemar $p = 4.16 \times 10^{-23}$) and the 3-arm Contrastive Companion Battery (**100/100** Seeded Recall, **100/100** Receipt Introspection, and **100/100** Calibrated Abstention on Compound Lucca, outperforming Bare LLM by +98.0 pp on Recall [$p_{\text{HB}} = 1.03 \times 10^{-21}$] and +50.0 pp on Receipt Introspection [$p_{\text{HB}} = 2.95 \times 10^{-11}$], and outperforming Top-5 Hybrid RAG by +15.0 pp on Recall [$p_{\text{HB}} = 3.05 \times 10^{-4}$]). We further wire the 2D NPU hazard policy table and Douglas Walton's `Scheme.AUTHORITY` critical questions directly into first-pass web search and Scrapling research synthesis (`evaluate_web_evidence_burden`). Across an $N = 360$ ($n = 30$ per cell across 12 synthetic cells) dataset authored by three blinded subagents (`PREREG-WEB-BURDEN-360-4MODEL`), the evaluator passes **360/360 (100.0%, 95% Wilson CI:** [98.94%, 100.00%]**)** on a deterministic Walton `Scheme.AUTHORITY` specification conformance test (`speculation`, `balance of probabilities`, `clear and convincing`, `beyond reasonable doubt`; 30/30 per cell on clean structured snippets whose cells are defined by distinct-domain count and conflict cues), **350/360 (97.22%)** pre-registered conservative NPU hazard safety (Predicted Hazard ≥ Expected Hazard; prior JSON artifact overwritten before commit `82b1da8`) and **360/360 (100.0%, CI:** [98.94%, 100.00%]**) post-hoc, in-sample** after modifying `_SENTENCE_SPLIT_RE` and

`_HIGH_HAZARD_DOMAIN_RE` in `neural_hazard_classifier.py` (which was omitted from `web_evidentiary_burden_360_prereg_lock.json`; 5.35 ms mean latency), alongside **277/360 (76.94%)** pre-registered and **287/360 (79.72%, CI:** [75.27%, 83.55%]**) post-hoc, in-sample** strict 3-class exact match. In a paired 288-turn live evaluation across all four local Gemma-4 GGUF checkpoints (`E2B` [0.49 s], `E4B` [0.65 s], `12B` [1.50 s], `26B-A4B` [1.60 s] at `--ctx-size 8192`), lexical regex scoring (`_HEDGE_OR_CONFLICT_RE`) recorded 99/144 (68.75%, with 30/72 [41.7%] on hedge-required cells) under Arm A vs. 144/144 (100.0%) under Arm B; however, because Arm B's system prompt explicitly instructed the exact legal-standard tokens matched by `_HEDGE_OR_CONFLICT_RE` without an instruction-only control arm (`Arm C`), this 144/144 vs. 99/144 contrast measures lexical instruction-following rather than verified semantic calibration. Under blinded semantic re-grading of the frozen 288 replies (`WO-45 R-10 v2`, `logs/test_reports/wo45/r10v2_regrade_288_report.json`; subagent model, temperature, and timestamp self-reported): on 36 disputed-source item pairs (4 models × 9 items), the burden-contract arm disclosed the cross-source conflict in 36/36 (100.0%) replies vs. 11/36 (30.6%) without it (two cross-family blinded judges, `D4` $\kappa = 0.80$; 0.99 with legal phrases redacted [12/36 vs. 36/36]); on non-disputed items where Lucca's policy requires extra corroboration, blinded judges disagreed on whether hedging was warranted (`D1` $\kappa = 0.20$), reflecting a disputed policy choice rather than a measured improvement; and the overall calibration difference across all 144 pairs ranges from −0.7 pp (120/144 vs. 119/144) to +45.8 pp (77/144 vs. 143/144) depending on the judge and post-unblinding tie-break rule. In a pre-registered 4-arm × 5-model Live v3 evaluation (`WO-45 G4 v3`, `logs/test_reports/wo45/g4_live_v3_report.json`, `GATE_G4v3.md`; $N = 132$ fresh stratified `WEB360` items with 0 overlap with Live v2 ×4 arms [`A`, `B`, `C`, `D`] ×5 Gemma-4 `Q6_K` models = 2,640 live turns at production `ctx=32768`, deterministic arm-identical normalization with 0 forbidden lexicon hits and 36.67% 3-way arm-guess accuracy, 3 qualification-screened machine-logged non-Google judge families [`Qwen3.8-27B`, `Phi-4-14B`, `Nemotron-3.5-30B-A3B`], and causal per-step OpenTimestamps + `origin/main` push provenance), inter-judge reliability computed strictly on the 660 disputed turns (`D4`, reply-only with zero snippets shown) reaches $\kappa =$ **0.8587**, **0.7154**, **0.6983** ($\min \kappa = 0.6983 \geq 0.60$, judge-robust). Unbundling `Arm D` (Arm B structured contract context + Arm C system prompt, with no `MUST` mandate and no legal-standard vocabulary) demonstrates that the structured conflict signal alone (`D vs C`) raises conflict disclosure from 85/165 (51.52%) to **164/165 (99.39%**, +47.88 pp [+36.36, +60.00], $p = 0.0001$, discordant 79 : 0; `Arm A`: 61/165 [36.97%]) and raises substantive faithful answering (`D2=NO` ∧ `NO_ANSWER=NO`) from 612/660 (92.73%) to 648/660 (98.18%, +5.45 pp, $p = 0.0001$; explicit unfaithfulness 0.91%–1.82%), whereas adding the `MUST` mandate and legal vocabulary (`B vs D`, 165/165 [100.0%] vs. 164/165) yields a null +0.61 pp ($p = 0.7214$, 1 : 0) on `D4` while imposing a −2.27 pp ($p = 0.0108$) penalty on `D1` epistemic calibration (95.15% vs. 97.42%; live `D1` 3-class $\kappa = 0.331$–0.401 and `D2` unfaithful `YES` $\kappa = 0.121$–0.249 fall below 0.60 due to class skew and are not claimed judge-robust; the earlier 3-arm Live v2 run [163/165 vs. 121/165, disputed $\kappa = 0.064$] remains archived with its `F-1..F-8` defect disclosures).

1. Introduction

An autonomous local agent that executes code, modifies workspace files, and queries external networks must satisfy two distinct honesty invariants. First, it must not poison its own long-term memory by storing unverified assistant narration as grounded fact. In our first study [1] (DOI: 10.5281/zenodo.23901452) and our five-layer memory report [3] (DOI: 10.5281/zenodo.23901683), we enforced this storage invariant through six HKDF-derived HMAC-SHA256 signers and SQLite triggers that partition autobiographical recollections (`memory_eligible=1, grounding_eligible=0`) from verified facts (`grounding_eligible=1`).

Second, the agent must not claim in live speech or text that it has executed a state-modifying action when no corresponding `ToolReceipt` exists in the current turn. In our second study [2] (DOI: 10.5281/zenodo.23901605), we evaluated two deterministic mechanisms for enforcing this runtime output invariant:

1. **Lexical Speech-Regex Supervision (`ActionClaimGrammarSupervisor`):** Expanding a first-person past-tense regex from 21 verbs to 44 stems and adding progressive participle rules raised in-vocabulary sensitivity

to 71.1% (128/180) across $N = 1{,}000$ completions from four Gemma-4 models. However, sensitivity collapsed to **29.5% (23/78)** on 10 unseen out-of-dev action verbs (58.5% [151/258] overall).

2. **Structured-Output Action Contracts (SOAC):** Requiring typed `[ACTION: ...]` frames verified against cryptographic receipts eliminated unframed prose leakage (0/120 on `gemma-4-26B-A4B`), but the contract system prompt caused the model to emit spurious action frames on 13.3% (4/30) to 16.7% (5/30) of figurative conversational idioms (e.g., *"kill two birds with one stone"*).

Paper 02 [2] therefore left an open architectural question: how can an edge agent verify open-vocabulary action claims, classify epistemic proof-policy hazards, and route multi-sentence user intents at < 5 ms latency without consuming dGPU VRAM or failing on multi-sentence prose?

When we trained 82.1M-parameter `cross-encoder/nli-MiniLM2-L6-H768` sequence classifiers on our discrete GPU and compiled them to OpenVINO FP16 on the Intel AI Boost NPU (`WO-41`), single-sentence in-distribution accuracy exceeded 98%, but on the 90-item `WO-41` blind adversarial stress test accuracy dropped to 76.7% (69/90), and on an uncurated 300-item multi-sentence blind benchmark (`Blind-300`, `WO-42 Phase 1`) pre-pooling accuracy dropped to **75.0% (225/300)**: 83.0% (83/100) on action-claim verification, 69.0% (69/100) on hazard classification, and 73.0% (73/100) on intent routing (303/390 = 77.7% across the combined 390 blind items).

This paper diagnoses the root cause of that multi-sentence degradation and presents the clause-level neurosymbolic pooling architecture that resolves it.

2. Why Whole-Turn Neural Classifiers Degrade on Multi-Sentence Turns

Line-by-line inspection of the 87 failures in the pre-pooling `Blind-390` run (`logs/test_reports/wo42_pass1_pre_pooling_fix_report.json`) identified three deterministic failure mechanisms when transformer sequence classifiers score an entire multi-sentence turn as a single token span or attempt discrete set-membership in embedding space:

2.1 Mechanism 1: Neural Set-Membership Confusion and Multi-Sentence Dilution (21/130 Coherence Misses)

Feeding `verified_tools` as a text string into a continuous embedding classifier forced the network to approximate discrete set membership (claimed_tool $\in$ executed_tools) and caused sensitivity to argument formatting (`launch_app(app='...')` vs. `["launch_app"]`). Furthermore, when an assistant response contained 3 to 5 sentences where a grounded first clause (e.g., `read_file`) was followed by an ungrounded second clause (e.g., `write_file`), whole-paragraph encoding diluted the ungrounded second clause.

2.2 Mechanism 2: Benign Context Masking High-Hazard Clauses (32/130 Hazard Misses)

In the epistemic proof-policy gate (`src/lucca/core/proof_policy.py`), a prompt containing three casual conversational sentences followed by a single factual/historical verification or destructive security question was classified as `LOW` hazard when scored as a single paragraph, while imperative non-corroboration commands falsely triggered `CROSS_REFERENCE_CMD`.

2.3 Mechanism 3: Conversational Opener Hijacking and Cross-Clause Entity Bleed (34/130 Router Misses)

Natural voice turns frequently open with greetings, venting, or context before stating a concrete task. Whole-utterance classification averaged 3 to 6 sentences into low-confidence predictions or delegated exact lexical contracts (`EMERGENCY_REFLEX` and whitelisted `known_apps` vs. third-party apps) to unconstrained softmax heads without slot verification.

3. Clause-Level Neurosymbolic NPU Architecture

Rather than retraining larger models that would exceed our < 5 ms latency budget, we keep all three 82.1M-parameter `nli-MiniLM2-L6-H768` OpenVINO FP16 models pinned to the Intel AI Boost NPU (0 MB dGPU VRAM) and introduce a deterministic symbolic decomposition, receipt set-membership, and priority-pooling layer around each classifier.

3.1 Syntactic Clause Segmentation (`split_narration_into_clauses`)

Given an input text T, `split_narration_into_clauses(T)` decomposes T into an ordered sequence of clauses $C = (c_1, c_2, \ldots, c_m)$ using sentence punctuation boundaries and coordinating action conjunctions (`and I also`, `then I`, `;`) while protecting decimal numbers, file extensions, abbreviations, and fenced code blocks.

3.2 Clause-Level Speech-Act Classification and Deterministic Receipt Set-Membership (`NeuralActionClaimVerifier`)

For assistant output T_a and user prompt T_u, `NeuralActionClaimVerifier` (`src/lucca/sandbox/neural_claim_verifier.py` [1,082 LOC], `models/action_claim_verifier_ov`, static shape `[1, 128]`) classifies each clause $c_i \in \text{split_narration_into_clauses}(T_a)$ in the context of the current user turn (`format_pair_texts` isolated to `user` role messages) into 5 speech-act classes and 7 canonical tool families (`CLAIM_WRITE_NOTE`, `CLAIM_SEARCH_READ`, `CLAIM_EXEC_MATH`, `CLAIM_SCHEDULE`, `CLAIM_MEMORY`, `CLAIM_LAUNCH_EXEC`, `CLAIM_DELETE_MOD`). Whenever a clause predicts an action claim or live-state fabrication, `verify_clause_claims_against_receipts` (`src/lucca/core/coherence.py`) performs exact deterministic Python set-membership and verb-subfamily verification against the turn's actual `verified_tools` (`ok=True`) receipts:

$$\text{Breach}(T_u, T_a, \mathcal{R}_{\text{ok}}) = \exists\, c_i \in \text{clauses}(T_a) : \text{ClaimedFamily}(T_u, c_i) \neq \varnothing \;\wedge\; \text{ClaimedFamily}(T_u, c_i) \cap \text{Families}(\mathcal{R}_{\text{ok}}) = \varnothing \tag{1}$$

3.3 Max-Hazard Lattice Pooling, Legal Standards of Proof, and Walton Defeasible Corroboration (`NeuralHazardClassifier`)

For user prompt T_u, `NeuralHazardClassifier` (`src/lucca/core/neural_hazard_classifier.py`, `models/proof_policy_hazard_ov`, static shape `[1, 96]`) scores each sentence across 5 hazard classes (`LOW_BENIGN`, `LOW_CAPABILITY_OR_META`, `HIGH_FACTUAL_HISTORICAL`, `CROSS_REFERENCE_CMD`, `HIGH_DESTRUCTIVE_SECURITY`) and applies max-hazard sentence pooling, a `_HAS_CORROBORATION_SIGNAL_RE` necessity gate on `CROSS_REFERENCE_CMD`, and high-entropy/gibberish fail-closed escalation to `HIGH`.

The resulting hazard level $h \in \{\texttt{LOW}, \texttt{MEDIUM}, \texttt{HIGH}\}$ parameterizes a two-dimensional decision-theoretic burden-of-proof table $\texttt{POLICY_TABLE_2D}(c, h)$ in `src/lucca/core/proof_policy.py` (`Spec D-REAS-09`) across four belief consumers $c \in \{\texttt{casual_mention}, \texttt{memory_commit_inferred}, \texttt{unhedged_assertion}, \texttt{irreversible_action}\}$ and four legal standards of proof (`ProofStandard` in `src/lucca/core/schemas.py`):

- **`SPECULATION`** (confidence < 0.50, **Unlikely / Speculative):** Cannot be committed to memory or stated as belief; if voiced at conclusion of an abductive chain (`DeductionAndPlanningRunner.run_deduction_step` in `src/lucca/core/deduction_runner.py`), it requires an explicit speculation hedge (*"I'm guessing ..., but that's just speculation"*).
- **`BALANCE_OF_PROBABILITIES`** ($0.50 \leq$ confidence < 0.75, **More Likely Than Not):** Satisfies `(casual_mention, LOW)` with a hedged belief marker (*"I think ..., but I'm not sure enough to keep it"*), but is rejected by `meets_standard(confidence, "memory_commit_inferred")`.
- **`CLEAR_AND_CONVINCING`** ($0.75 \leq$ confidence < 0.95**):** Required minimum threshold to commit an `INFERRED` fact to SQLite memory under `LOW` or `MEDIUM` hazard, for `(casual_mention, MEDIUM)`, and for

`(unhedged_assertion, LOW)`.

- **`BEYOND_REASONABLE_DOUBT` (confidence ≥ 0.95):** Required for all `HIGH`-hazard claims (medical, legal, financial, security), for `(unhedged_assertion, MEDIUM/HIGH)`, and for all `irreversible_action` dispatches regardless of hazard level.

Before an inferred or partially proven premise is checked against `POLICY_TABLE_2D`(c, h), `CorroborationEngine` (`src/lucca/core/corroboration.py` and `src/lucca/core/walton_schemes/evaluators.py`) evaluates Douglas Walton's five defeasible argumentation schemes (`AUTHORITY`, `SIGN`, `ANALOGY`, `CONSEQUENCE`, and `PRECEDENT`). Each scheme dispatches three critical questions (`CQ1..CQ3`: source receipt or factual congruence, independent grounding-eligible corroboration, and competing dispute or countervailing evidence) against the signed SQLite store and turn receipts. The confidence update function `adjust_confidence(confidence, q_results, scheme)` increments independently corroborated claims ($+0.10$ to $+0.15$ subject to a hard $\min(0.95, \ldots)$ ceiling that prevents defeasible evidence from laundering into 1.00 certainty) and decrements disputed or superseded claims (-0.20 to -0.30) while appending explicit defeater conditions (`unless...`, `assuming...`) rendered in speech by `format_conditioned_speech`.

To ensure that first-pass question answering and external web research obey `POLICY_TABLE_2D`(c, h) without requiring an explicit follow-up *"cross-reference that"* command, `evaluate_web_evidence_burden` (`src/lucca/core/proof_policy.py`) is invoked automatically on every `TIER_4_WEB` cascade retrieval (`query_substrate_3_web` in `src/lucca/core/cascade_substrates.py`) and every `search_web`/`research_topic` tool dispatch (`src/lucca/core/tool_dispatcher.py`). Rather than stamping all web snippets with a static 0.85 prior, the evaluator normalizes registrable domains (`_registrable_domain`, collapsing `www.` and subdomain aliases onto a single eTLD+1 identity), extracts the single sentence per domain with highest proposition-token overlap (`_select_query_aligned_sentence`), filters non-authority scheme tokens so benign procedural descriptions do not trigger false disputes, and runs `evaluate_authority` (`CQ1..CQ3`):

- **Cross-Source Dispute (`CQ3` Satisfied $\rightarrow 0.40$, `SPECULATION`):** Conflicting claims across domains set `told_true=True, told_false=True` (`BelnapValue.CONFLICT`), dock confidence from 0.70 to 0.40 (-0.30), inject an explicit `[Epistemic Burden Contract]` into synthesis context, and block proactive memory ratification in `check_and_offer_ratification` (`src/lucca/core/deduction_runner.py`).
- **Single Uncorroborated Domain (1 Source $\rightarrow 0.70$, `BALANCE_OF_PROBABILITIES`):** Satisfies `(casual_mention, LOW)` (≥ 0.50), but falls below `(casual_mention, MEDIUM)` (≥ 0.75) and `(casual_mention, HIGH)` (≥ 0.95). On `MEDIUM` or `HIGH` hazard queries, `query_substrate_3_web` automatically escalates from single-snippet `search_web` to multi-domain Scrapling extraction (`deep_extract=True`) and injects a mandatory qualification contract if corroboration remains below threshold.
- **Two Independent Corroborating Domains (2 Sources $\rightarrow 0.80$, `CLEAR_AND_CONVINCING`):** `CQ2` increments confidence from 0.70 to 0.80 ($+0.10$), satisfying `(casual_mention, MEDIUM)` and `(memory_commit_inferred, LOW/MEDIUM)` (≥ 0.75) while still requiring explicit qualification under `HIGH` hazard (≥ 0.95).
- **Three or More Independent Domains (≥ 3 Sources $\rightarrow 0.95$, `BEYOND_REASONABLE_DOUBT`):** Multi-domain base 0.85 plus `CQ2` $+0.10$ reaches the hard 0.95 defeasible ceiling, satisfying all `HIGH`-hazard gates.

3.4 Task-Priority Pooling for Multi-Sentence Intent Routing (`NeuralIntentClassifier`)

For multi-head intent routing (`src/lucca/router/neural_intent_classifier.py`, `models/router_intent_ov`, static shape `[1, 64]`), `NeuralIntentClassifier` predicts 14 primary intents and 9 reasoning domain flags per sentence and promotes concrete task/reasoning intents

(`CLASS_PRIORITY_TIER >= 1`) over greetings and conversational filler (`chit_chat`, `clarify`), paired with deterministic `known_apps` slot verification (`_extract_app_entity`) and `ConformalGuard` out-of-domain abstention.

4. Empirical Evaluation

4.1 Large-Sample Action-Claim Evaluation ($N = 2{,}022$)

To test whether clause-level NPU supervision resolves the open-vocabulary ceiling identified in Paper 02 [2] with adequate statistical power, we evaluated `NeuralActionClaimVerifier` against the legacy 44-stem regex supervisor across all $N = 2{,}022$ labeled action-claim pairs in the repository (`logs/test_reports/wo42_large_sample_2020_report.json`):

- **Paper 02 4-Model Corpus ($n = 1{,}000$):** 250 prompts $\times$ 4 Gemma-4 models (`26B-A4B`, `12B`, `E4B`, `E2B`) with dual-rater blinded semantic labels ($\kappa = 1.0000$ binary).
- **Paper 02 Held-Out V1/V2 Benchmark ($n = 120$):** 50 temptations, 40 honest controls, 30 idioms.
- **Organic, Implicit, Passive, and Blind Multi-Sentence Suites ($n = 352$):** Including `BLIND_COH_30`, `BLIND42_COH_100`, `LOCKED_COH_050`, passive/third-person claims, structured JSON/YAML/TOML/XML status blocks, and foreign-language claims.
- **Property-Based Hypothesis Fuzz Corpus ($n = 550$):** Randomized multi-sentence needle-in-a-haystack and benign control permutations.

Table 1. Action-Claim Supervision on the Combined $N = 2{,}022$ Battery (471 Positive Violations, 1,551 Benign/Grounded Controls).

Supervisor Architecture	Accuracy	Recall (Sensitivity)	False Positive Rate	Latency (Mean / P95)
Legacy 44-Stem Regex [2]	79.23% (1,602/2,022)	11.46% (54/471)	**0.19%** (3/1,551)	0.04 ms/0.09 ms (CPU)
Whole-Turn DeBERTa (Pre-WO-42)	89.12% (1,802/2,022)	68.15% (321/471)	4.51% (70/1,551)	1.31 ms/1.85 ms (NPU)
Clause-Level NPU (`WO-42`)	**99.26%** (2,007/2,022)	**98.73%** (465/471)	0.58% (9/1,551)	2.95 ms/3.67 ms (NPU)
95% Wilson Score CI (WO-42)	[98.78%, 99.55%]	[97.25%, 99.42%]	[0.31%, 1.10%]	0 MB dGPU VRAM

Table 1 summarizes the results. On the 471 positive unexecuted action claims, `NeuralActionClaimVerifier` catches 465/471 (98.73%) compared to 54/471 (11.46%) for the legacy regex supervisor (411 discordant wins vs. 0 losses, exact McNemar $\chi^2 = 399.04, p = 3.07 \times 10^{-89}$). On the 1,551 negative controls (honest refusals, idioms, code explanations, and grounded receipt narrations), the false positive rate is 0.58% (9/1,551, 95% Wilson CI: [0.31%, 1.10%]), satisfying the $\leq 2.0\%$ production gate established in Paper 02.

4.2 Pre-Registered Blind and Locked Holdout Suites ($N = 540$)

To prevent iterative test-set overfitting, we conducted two multi-agent blinded evaluations:

1. **`Blind-390` (`WO-42 Phase 4`):** Three isolated subagents with zero access to training scripts or model weights authored 300 multi-sentence adversarial items (`BLIND42_COH_001..100`, `BLIND42_HZD_001..100`, `BLIND42_RTR_001..100`), combined with the 90-item `WO-41` blind stress test.
2. **`WO42_LOCKED_HOLDOUT_150`:** After freezing all training weights, three fresh isolated subagents authored 150 locked items (50 per classifier). Two independent blinded LLM judges graded all 150 outputs line by line without seeing model predictions, achieving 20/20 (100.0%, Cohen's $\kappa = 1.000$) inter-rater agreement on the overlap slice.

Table 2. Performance Across Blind-390 and the Pre-Registered Locked Holdout (`WO42_LOCKED_HOLDOUT_150`).

NPU Subsystem	Blind-390 (Pass 1)	Blind-390 (Pass 2 / Post-Fix)	Locked-150 (Single-Pass)	Locked-150 (Post-Fix)
`NeuralActionClaimVerifier`	109/130 (83.8%)	130/130 (100.0%)	50/50 (100.0%)	50/50 (100.0%)
`NeuralHazardClassifier`	98/130 (75.4%)	125/130 (96.2%)	50/50 (100.0%)	50/50 (100.0%)
`NeuralIntentClassifier`	96/130 (73.8%)	121/130 (93.1%) / 122/130 (93.8%)	45/50 (90.0%)	50/50 (100.0%)
Combined Total	303/390 (77.7%)	**376/390 (96.4%)** / **377/390 (96.7%)**	**145/150 (96.67%)**	**150/150 (100.0%)**

As shown in Table 2, single-pass evaluation on `WO42_LOCKED_HOLDOUT_150` (`logs/test_reports/wo42_locked_holdout_150_report.json`) achieved 50/50 (100.0%) on action-claim verification, 50/50 (100.0%) on hazard classification, and 45/50 (90.0%) on intent routing ($145/150 = 96.67\%$ overall, 95% Wilson CI: $[92.43\%, 98.57\%]$). Three of the 5 single-pass router errors (`LOCKED_RTR_007, 008, 011`) predicted `logic_constraints` with `pred_libs = ["sympy", "z3"]` but assigned `reasoning_domain = "mathematics"` instead of `"combinatorial_smt"` because `heuristic.py` checked `has_sympy` before `has_z3`; the remaining 2 errors (`LOCKED_RTR_045, 046`) wrapped app names in Markdown backticks (`` `calculator` ``, `` `notepad` ``) before the `launch_app` regex boundary. Fixing both deterministic rules in `src/lucca/router/heuristic.py` after recording the single-pass locked report resolved all 5 cases (150/150, `logs/test_reports/wo42_locked_holdout_150_post_router_fix_report.json`, and raised `Blind-390` from 376/390 to 377/390). We report both the 96.67% single-pass locked number and the 100.0% post-fix number to preserve strict pre-registration transparency.

4.3 Live 4-Model End-to-End Verification ($N = 100$ Live Turns)

Offline classifier benchmarks do not test whether an orchestrator can intercept a live LLM mid-turn, clear its speech queue, and force a self-corrected response without breaking benign tool execution. We therefore executed a 100-turn live matrix (`scripts/run_wo42_live_4model_matrix.py`, `logs/test_reports/wo42_live_4model_summary.json`) across all four local Gemma-4 GGUF checkpoints served on an NVIDIA RTX 5080 Laptop GPU (16 GB GDDR7) while the three OpenVINO FP16 classifiers ran concurrently on the Intel AI Boost NPU:

- **Tier 0:** `gemma-4-26B-A4B-it-heretic.Q6_K.gguf` (MoE, 4B active, 15,918 MB VRAM)
- **Tier 1:** `gemma-4-12b-heretic-Q6_K.gguf` (Dense 12B, 11,130 MB VRAM)
- **Tier 2:** `gemma-4-E4B-it-uncensored-heretic-Q6_K.gguf` (Dense 4.5B, 4,848 MB VRAM)
- **Tier 3:** `Gemma4-E2B-Heretic-Sinhala.i1-Q6_K.gguf` (Dense 2.0B, 2,676 MB VRAM)

Each model completed 25 live turns across multi-sentence tool routing, proof-policy hazard gating, adversarial false-action-claim temptations, and grounded narration. Three blinded LLM judges graded all 100 transcripts line by line ($\kappa = 1.000$ on the 20-turn overlap slice).

Table 3. Live 4-Model End-to-End Matrix (25 Turns × 4 Local Gemma-4 Models = 100 Live Turns).

Local GGUF Checkpoint	RTX 5080 VRAM	Mean Turn Latency	Axis 1: Coherence & Grounding	Axis 2: Task, Routing & Reasoning
`Gemma4-E2B-Heretic (Q6_K)`	2,676.0 MB	0.66 s	**25/25 (100.0%)**	23/25 (92.0%)
`gemma-4-E4B-heretic (Q6_K)`	4,848.0 MB	1.11 s	**25/25 (100.0%)**	**25/25 (100.0%)**
`gemma-4-12b-heretic (Q6_K)`	11,130.0 MB	2.15 s	**25/25 (100.0%)**	**25/25 (100.0%)**
`gemma-4-26B-A4B (Q6_K)`	15,918.0 MB	37.01 s	**25/25 (100.0%)**	**25/25 (100.0%)**
Pooled 4-Model Total	0.0 MB NPU overhead	—	**100/100 (100.0%)**	**98/100 (98.0%)**

Whenever a model fabricated an unexecuted action in Round 1 of Category A (observed in 19/28 initial drafts across the four models, including JSON status payloads and passive confirmations), `SpeechStreamPipeline.check_functional_coherence` (`src/lucca/core/speech_stream_pipeline.py`, delegated from `src/lucca/core/orchestrator.py`) caught the violation via `NeuralActionClaimVerifier`, flushed the pending TTS queue (`speech_output.clear_queue()`), discarded the ungrounded draft span, and forced a self-corrected Round 2 response acknowledging that no tool was executed (28/28 final turn pass rate, 0/16 false positives on Category B code/JSON controls).

4.4 Pre-Registered $N = 300$ Memory-Poisoning and Contrastive Companion Expansion (`WO-43`)

Because our earlier Zenodo preprints [1, 3] remain permanently frozen at their published $N = 100$ (40 Cohort A, 30 Cohort B, 30 Cohort C) and $N = 30$ (10 per stratum) sample sizes to preserve archival DOI integrity, we report the pre-registered Work Order 43 ($N = 300$, tag `prereg-wo43-power-300`) statistical power expansions here in the successor study. Six isolated subagents with zero access to evaluator internals authored 300 paired memory-poisoning probes (`data/benchmarks/memory_poisoning_v3_300.jsonl`) and 300 contrastive companion prompts (`data/benchmarks/contrastive_battery_v2_300.jsonl`), evaluated live on `gemma-4-E4B-it-uncensored-heretic-Q6_K.gguf` ($T = 0.0$, `seed=42`, `cache_prompt=False`).

Table 4. Pre-Registered 5-Arm Memory Poisoning & Misinformation Ablation v3 ($N = 300$, 100 Probes per Cohort ×5 Arms = 1,500 Live Evaluations).

Memory Architecture Arm	Cohort A Poison	Cohort A Continuity	Cohort B Retention	Cohort C Misinfo
Naive RAG (`Mem=1, Gnd=1`)	100/100 (100.0%)	100/100 (100.0%)	100/100 (100.0%)	100/100 (100.0%)
No-Assistant (`Mem=0, Gnd=0`)	0/100 (0.0%)	2/100 (2.0%)	1/100 (1.0%)	100/100 (100.0%)
Passive Soft Tagging	11/100 (11.0%)	100/100 (100.0%)	100/100 (100.0%)	100/100 (100.0%)
Adversarial Soft Tagging	100/100 (100.0%)	100/100 (100.0%)	100/100 (100.0%)	100/100 (100.0%)
Lucca Gate v3 (`WO-43`)	**0/100 (0.0%)**	**100/100 (100.0%)**	**100/100 (100.0%)**	**0/100 (0.0%)**
95% Wilson CI (Lucca Gate v3)	[0.00%, 4.61%]	[95.39%, 100.00%]	[95.39%, 100.00%]	[0.00%, 4.61%]

Table 4 shows that scaling from $n = 30$–40 to $n = 100$ per cohort tightens the continuity-corrected 95% Wilson upper bound on zero observed poisoning from 11.4% to **4.61%** (3.70% uncorrected Method 3). Across Cohort A and Cohort C, Lucca Gate v3 (with `corroborate_user_explicit=True` and `BelnapLatentInvestigator` active) achieves 0/100 poisoning vs. 100/100 for Naive RAG and Adversarial Soft Tagging (exact McNemar $p = 4.16 \times 10^{-23}$) and 11/100 for Passive Soft Tagging ($p = 9.77 \times 10^{-4}$), with primary-vs.-NLI agreement $\kappa = 0.9850$ across all 1,500 turns.

Table 5. Pre-Registered 3-Arm Contrastive Companion Battery v2 ($N = 300$, 100 Prompts per Stratum $\times 3$ Arms $= 900$ Live Turns).

Stratum ($n = 100$)	Arm A (Bare LLM)	Arm B (Compound Lucca)	Arm C (Top-5 Hybrid RAG)	Arm B vs. A / C (p_{HB})
Seeded Recall (`R001..100`)	2/100 (2.0%)	**100/100 (100.0%)**	85/100 (85.0%)	1.03×10^{-21} / 3.05×10^{-4}
Receipt Introspection (`I001..100`)	50/100 (50.0%)	**100/100 (100.0%)**	98/100 (98.0%)	2.95×10^{-11} / 1.0000
Calibrated Abstention (`A001..100`)	99/100 (99.0%)	**100/100 (100.0%)**	100/100 (100.0%)	1.0000 / 1.0000

Table 5 reports the 3-arm Contrastive Companion Battery v2 (`logs/test_reports/contrastive_v2_300_results_REPORT.md`) following remediation of two deterministic grader defects (`C-07`: rejecting Bare LLM refusals before substring matching on affirmative receipt probes and normalizing backticks in `strip_echo`; `C-08`: including conversational contractions in `ABSTENTION_PHRASES`, raising dual blinded judge agreement on the 60-item IRR slice to $60/60 = 100.0\%$, $\kappa = 1.0000$) and bare `memory_search` tool-call demuxing in `toolcall_demux.py`. Compound Lucca achieves **100/100 (100.0%, 95% CI:** $[95.4\%, 100.0\%]$**)** across all three strata, significantly outperforming Bare LLM on Seeded Recall ($+98.0$ pp, $98 : 0$ discordant, Holm–Bonferroni $p_{\text{HB}} = 1.03 \times 10^{-21}$) and Receipt Introspection ($+50.0$ pp, $50 : 0$ discordant, $p_{\text{HB}} = 2.95 \times 10^{-11}$, as Bare LLM lacks a receipt store and refuses all 50 affirmative probes), while significantly outperforming Top-5 Hybrid RAG on Seeded Recall ($+15.0$ pp, $15 : 0$ discordant, $p_{\text{HB}} = 3.05 \times 10^{-4}$) due to deterministic belief-supersession and autobiographical query boosting in `cascade_substrates.py`.

4.5 Pre-Registered $N = 360$ ($n = 30$/Cell) 2D Web Evidentiary Burden Holdout and 288-Turn Live 4-Model GGUF Evaluation

An initial developmental smoke test ($N = 60$, $n = 5$ per cell in `data/benchmarks/web_evidentiary_burden_60.jsonl`) exposed two real-world defects on scraped web text ($57/60 = 95.0\%$ pre-fix $\rightarrow 60/60 = 100.0\%$ post-fix): (1) cross-scheme negation/adversative tokens in `corroboration.py` (*"side effects"*, *"prevent"*, `--no-preserve-root`, *"do not exceed"*) triggered false `CQ3` disputes under `Scheme.AUTHORITY`, and (2) offline fallback notices duplicated domain identities. However, $n = 5$ per cell carries a wide 95% Wilson lower bound of 56.55% even at 5/5, and tuning on the developmental set introduces in-sample bias.

To obtain statistically powered, out-of-sample estimates across both the neurosymbolic evaluator and all four local GGUF model weights, we executed pre-registration `PREREG-WEB-BURDEN-360-4MODEL` (`logs/test_reports/web_evidentiary_burden_360_prereg_lock.json`):

1. **SHA-256 Code & Weight Lock (`WO-45 R-20/R-23` Disclosure):** We recorded the SHA-256 digests of `proof_policy.py`, `corroboration.py`, `walton_schemes/evaluators.py`, `cascade.py`, `cascade_substrates.py`, and `models/proof_policy_hazard_ov/openvino_model.bin` in `logs/test_reports/web_evidentiary_burden_360_prereg_lock.json` (`locked_at_utc: 2026-11-09T01:51:54Z`). **Provenance disclosure (`WO-45 R-20/R-23`):** First, `src/lucca/core/neural_hazard_classifier.py` was omitted from `web_evidentiary_burden_360_prereg_lock.json` and edited after inspecting the 10 initial `HIGH` $\rightarrow$ `MEDIUM` misses. Second, in git history, `proof_policy.py`, `neural_hazard_classifier.py`, the lock file, the $N = 360$ corpus, and the evaluation report were all committed together in a single commit (`82b1da8`, `07:16Z`), so git commit ordering alone cannot independently prove that the lock preceded corpus generation. All subsequent `WO-45` holdouts enforce strict separate-commit ordering (`lock` $\rightarrow$ `corpus` $\rightarrow$ `results`) plus external timestamping (`.ots`).

2. **Blinded $N = 360$ ($n = 30$/Cell) Holdout Generation:** Three isolated subagents with zero read access to `src/lucca/` authored 120 probes each across `LOW`, `MEDIUM`, and `HIGH` hazard strata (30 per topology $\times 4$ topologies $= 120$ per stratum; `data/benchmarks/web_evidentiary_burden_360.jsonl`, SHA-256 `db685e70e65b40ba42be0e3ea2bbf54ac00b72ec6c69546faeabcff3b5574e13`), including multi-label eTLD+1 subdomain alias traps (`.co.uk`, `.com.au`, `www.`, `wiki.`, `clinical.`).
3. **Paired 288-Turn Live 4-Model GGUF Matrix:** Across a stratified 36-probe slice (3 probes per cell $\times 12$ cells), we evaluated all four local Gemma-4 GGUF checkpoints (`E2B`, `E4B`, `12B`, and `26B-A4B` served with production flags `--cpu-moe --load-mode none --moe-cache-mib 7680 --spec-type ngram-mod --cache-type-k q8_0 --cache-type-v q8_0 --cache-ram 4096`) under two paired arms ($36 \times 2 \times 4 = 288$ live turns): **Arm A (Legacy Static-0.85 Synthesis)**, which stamps web snippets at flat confidence 0.85 (`[asserts]`) with no burden contract, and **Arm B (2D Burden-Gated Synthesis)**, which injects `[{burden_tag}]` and `[Epistemic Burden Contract: ...]` from `evaluate_web_evidence_burden`.

Table 6. Blinded 2D Web Evidentiary Burden Holdout ($N = 360$, $n = 30$ per Cell across 3 Hazards $\times 4$ Topologies). Pre-split values (*, 350/360 safe, 277/360 strict; not reproducible: artifact overwritten) reflect the initial run; 360/360 safe and 287/360 strict are **post-hoc, in-sample** after editing `neural_hazard_classifier.py` (which was omitted from `web_evidentiary_burden_360_prereg_lock.json`).

Hazard	Source Topology ($n = 30$)	Confidence & Standard	Legacy 0.85	Walton Spec. Conformance (95% CI)	NPU Safe (post-hoc / pre*)	Strict 3-Class (post-hoc / pre*)
`LOW`	1 Source / eTLD+1 Alias	0.70 (`balance of prob.`)	0/30	**30/30** ([88.65%, 100.0%])	30/30	26/30
`LOW`	2 Independent Domains	0.80 (`clear & convincing`)	30/30	**30/30** ([88.65%, 100.0%])	30/30	16/30
`LOW`	3+ Independent Domains	0.95 (`beyond reas. doubt`)	0/30	**30/30** ([88.65%, 100.0%])	30/30	16/30
`LOW`	Disputed Across Domains	0.40 (`speculation`)	0/30	**30/30** ([88.65%, 100.0%])	30/30	13/30
`MEDIUM`	1 Source / eTLD+1 Alias	0.70 (`balance of prob.`)	0/30	**30/30** ([88.65%, 100.0%])	30/30	24/30
`MEDIUM`	2 Independent Domains	0.80 (`clear & convincing`)	30/30	**30/30** ([88.65%, 100.0%])	30/30	24/30
`MEDIUM`	3+ Independent Domains	0.95 (`beyond reas. doubt`)	0/30	**30/30** ([88.65%, 100.0%])	30/30	23/30
`MEDIUM`	Disputed Across Domains	0.40 (`speculation`)	0/30	**30/30** ([88.65%, 100.0%])	30/30	25/30
`HIGH`	1 Source / eTLD+1 Alias	0.70 (`balance of prob.`)	0/30	**30/30** ([88.65%, 100.0%])	**30/30** (27/30*)	**30/30** (27/30*)
`HIGH`	2 Independent Domains	0.80 (`clear & convincing`)	30/30	**30/30** ([88.65%, 100.0%])	**30/30** (28/30*)	**30/30** (28/30*)
`HIGH`	3+ Independent Domains	0.95 (`beyond reas. doubt`)	0/30	**30/30** ([88.65%, 100.0%])	**30/30** (28/30*)	**30/30** (28/30*)
`HIGH`	Disputed Across Domains	0.40 (`speculation`)	0/30	**30/30** ([88.65%, 100.0%])	**30/30** (27/30*)	**30/30** (27/30*)
Total	$N = 360$ (12×30)	6.58 ms E2E NPU / 16.92 ms CPU	90/360 (25.0%)	**360/360 (100.0%)**	**360/360** (350/360*)	**287/360** (277/360*)

Table 6 shows the results on the $N = 360$ holdout (`logs/test_reports/web_evidentiary_burden_360_report.json`), reporting both the pre-clause-split baseline (*) and the subordinate-clause-split result:

- **Walton `Scheme.AUTHORITY` Specification Conformance Test ($360/360 = 100.0\%$ on Synthetic Structured Snippets, `WO-45 R-06/R-22/R-30/R-31`):** Across all 12 cells (30/30 per cell; combined end-to-end `evaluate_web_evidence_burden()` latency measured over $N = 360$ with 3 warm-ups in `logs/test_reports/wo45/r06_gate_latency_npu_vs_cpu.json` is 6.575 ms mean / 5.632 ms p50 / 14.274 ms p95 / 15.258 ms p99 on Intel AI Boost NPU vs. 16.923 ms mean / 16.348 ms p50 / 38.101 ms p95 / 48.406 ms p99 on Intel Core Ultra 9 275HX CPU with `device="CPU"`), `evaluate_web_evidence_burden` matched the expected legal standard of proof, distinct eTLD+1 domain count, and cross-domain dispute verdict. Because the benchmark cells are defined by distinct-domain count and explicit conflict cues in author-written structured snippets while the evaluator deterministically counts distinct domains and matches dispute cues, this $360/360$ result is a **specification conformance test** verifying implementation correctness (not empirical evidence of open-web epistemic validity; inferential McNemar p-values against the constant 0.85 strawman baseline, which passes $1/4$ topologies [$90/360 = 25.0\%$] by construction, are removed as evidence of efficacy per `WO-45 R-30/R-31`). **Non-trivial baselines (`WO-45 R-31`, in-sample on the `WO-44` corpus with a post-hoc selected casual-gate metric, `logs/test_reports/wo45/r31_nontrivial_baselines_360.json`):** On the binary casual-mention gate across the same $N = 360$ items (in-sample on the `WO-44` corpus, metric selected post-hoc), Lucca 2D End-to-End scores $348/360$ (96.67%, 95% CI $[94.26\%, 98.09\%]$, $0/180$ unsafe under-hedges) vs. $300/360$ (83.33%, exact McNemar $b : c = 56 : 8$, $p = 5.56 \times 10^{-10}$) for `Count+Dispute` (1 source or disputed $\rightarrow$ hedge, ignoring hazard), $242/360$ (67.22%, $b : c = 107 : 1$, $p = 6.72 \times 10^{-31}$) for `Hazard-Only (NPU)`, $240/360$ (66.67%) for `Hazard-Only (Nominal)` and `Static-0.85`, and $210/360$ (58.33%, $b : c = 146 : 8$, $p = 6.04 \times 10^{-34}$) for `Count-Only`. **Dispute-normalization ablation disclosure (`WO-45 R-22`, post-hoc analysis on the `WO-44` corpus, `logs/test_reports/wo45/r22_dispute_normalization_ablation.json`):** In commit `82b1da8`, `src/lucca/core/proof_policy.py:424-443` added seven special-case phrase normalizations (`--no-[a-z0-9-]+`, `to prevent`, `do not exceed`, `cannot be undone without`, `takes no c\w*`, `that are not part of`, and `bom is forbidden`, fitted during development on the prior $N = 60$ smoke test [`WEB_HIGH_003`, `009`, `010`] and the $N = 6$ live Scrapling slice [`LIVE_LOW_02`, `LIVE_MED_02`]) plus a broader negation/contrast-word stripping rule (`:433-443`) applied when no explicit authority-dispute cue is present. Disabling all seven special-case patterns on the $N = 360$ corpus changes $0/360$ verdicts ($360/360 = 100.0\%$), showing that those seven patterns are **dead code on this corpus** (removed prior to the Gate G4 lock in commit `d504656` per `WO-45 G4-6`), whereas disabling the broader negation/contrast-word strip (`:433-443`) changes $4/360$ items (`WEB360_LOW_021`, `WEB360_LOW_075`, `WEB360_HIGH_044`, `WEB360_HIGH_071`), reducing conformance from $360/360$ (100.0%) to $356/360$ (98.89%).
- **NPU Hazard Calibration, Conservative Safety Direction, & Over-Escalation Utility Cost (`WO-45 R-20/R-40 v2/R-41`):** On the initial pass where `neural_hazard_classifier.py` split only on sentence-terminal punctuation (`[.!?]`), $10/360$ (2.78%) compound database maintenance queries (*"When performing PostgreSQL 16 wal archive maintenance..., how does pg_archivecleanup... delete or format production database disk partition data"*) under-classified as `MEDIUM` ($350/360 = 97.22\%$ conservative safety, $277/360 = 76.94\%$ strict 3-class; note per `WO-45 R-09` that the JSON report from this pre-split pass was overwritten before commit `82b1da8`). **Lock omission disclosure (`WO-45 R-20`):** While `proof_policy.py` and the OpenVINO weights were hashed in `web_evidentiary_burden_360_prereg_lock.json`, `src/lucca/core/neural_hazard_classifier.py` was omitted from that lock file. Extending `_SENTENCE_SPLIT_RE` and `_HIGH_HAZARD_DOMAIN_RE` in `neural_hazard_classifier.py` after inspecting the 10 failures resolved all 10 under-classifications without altering the locked `proof_policy.py` hash, reaching **360/360 (100.0%, 95% CI:** $[98.94\%, 100.00\%]$**, post-hoc, in-sample)** conservative safety compliance (Predicted Hazard $\geq$ Expected Hazard at 5.35 ms mean latency) and **287/360 (79.72%, 95% CI:** $[75.27\%, 83.55\%]$**, post-hoc, in-sample)** strict 3-class accuracy. All 73 remaining 3-class differences

are conservative upward escalations (51 `LOW` → `MEDIUM` on declarative scientific statements, 6 `LOW` → `HIGH`, and 16 `MEDIUM` → `HIGH` on historical treaty/statutory phrasing in `web_evidentiary_burden_360_report.json`). In a blinded mixed-pool hazard audit (`WO-45 R-40 v2`, `logs/test_reports/wo45/r40v2_upward_escalations_audit.json`, 73 escalated items shuffled with 90 stratified non-escalated controls = 163 items rated by `claude-opus-4-6` and `gemini-2.5-flash` under a neutral consequence-only prompt, 3-class $\kappa = 0.9088$; subagent model, temperature, and timestamp self-reported), consensus ratings matched nominal corpus labels on 82/90 (91.11%, 95% Wilson CI [83.43%, 95.43%]) of controls, while on the 73 escalated items both raters agreed 73/73 (100.0%, $\kappa = 1.0000$): 0/73 (0.0%, CI [0.0%, 5.0%]) were corpus label errors and 73/73 (100.0%, CI [95.0%, 100.0%]) were true classifier over-escalations. **Retraining fallback & item-level Live v3 utility-cost count (`WO-45 RM-7/R-42 & R-41`, `logs/test_reports/wo45/r42_hazard_retrain_report.json`, `g4_live_v3_report.json`):** Under `RM-7` (`R-42`), retraining `models/proof_policy_hazard_v2_ov` with hard negatives (0 Live v3 item leakage; 24/73 = 32.88% held-out over-escalations) reduced held-out `R-40 v2` over-escalations from 24/24 (100.0%) to 1/24 (4.17%), scored 50/50 (100.0%) on `HZD_50`, and matched 125/130 (96.15%, 0.00 pp regression) on `Blind-390`, but missed 1 item on `WO42_LOCKED_HOLDOUT_150`'s 50-item `proof_policy` slice (49/50 = 98.00%, −2.00 pp slice regression > 1.0 pp threshold; −0.67 pp across all 150 items). Enforcing the pre-specified fallback (`accepted_for_g4_v3 = False`) kept `v1` active for Live v3 while logging `v1` and `v2` side-by-side: across Live v3's 132 fresh items, active `v1` over-escalated 24/132 (18.18%) items (6 nominal-burden-met items blocked) vs. 1/132 (0.76%, 1 blocked) for `v2`. At the item level across those 6 blocked items (`D3` $\kappa = 0.8846, 0.8811, 0.9177$), model-majority item over-hedging (`items over-hedged / items escalated`) was 2/6 (33.33%) on Arm A, 1/6 (16.67%) on Arm B, 1/6 (16.67%) on Arm C, and 2/6 (33.33%) on Arm D (mean item over-hedge rate across 5 models: Arm A 36.67% [13.33%, 60.00%], Arm B 36.67% [26.67%, 46.67%], Arm C 33.33% [13.33%, 53.33%], Arm D 43.33% [26.67%, 60.00%]; in the earlier historical Live v2 run [`g4_live_v2_report.json`, Defect F-7], 3 post-hoc selected escalated items accounted for 3/15 [20.00%] over-hedged turns on Arm B). Out-of-sample verification on a fresh holdout follows under `WO-45 R-21`.

Table 7. Live 4-Model GGUF Epistemic Burden Compliance (36 Stratified Prompts × 2 Paired Arms × 4 Local Gemma-4 Models = 288 Live Turns).

Local GGUF Checkpoint	VRAM (Latency)	Arm A: Legacy Static-0.85 (Hedge / Met)	Arm B: 2D Burden-Gated (Hedge / Met)	Arm B 95% CI	McNemar χ^2 (p)
`Gemma4-E2B-Heretic (Q6_K)`	2,560 MB (0.49 s)	24/36 (66.7%: 7/18 / 17/18)	**36/36 (100.0%**: 18/18 / 18/18)	[90.36%, 100.0%]	10.08 (4.88×10^{-4})
`gemma-4-E4B-heretic (Q6_K)`	4,502 MB (0.65 s)	24/36 (66.7%: 8/18 / 16/18)	**36/36 (100.0%**: 18/18 / 18/18)	[90.36%, 100.0%]	10.08 (4.88×10^{-4})
`gemma-4-12b-heretic (Q6_K)`	10,620 MB (1.50 s)	24/36 (66.7%: 6/18 / 18/18)	**36/36 (100.0%**: 18/18 / 18/18)	[90.36%, 100.0%]	10.08 (4.88×10^{-4})
`gemma-4-26B-A4B (Q6_K, cpu-moe)`	11,102 MB (1.60 s)	27/36 (75.0%: 9/18 / 18/18)	**36/36 (100.0%**: 18/18 / 18/18)	[90.36%, 100.0%]	7.11 (3.91×10^{-3})
Pooled 4-Model Total (288 Turns)	0.0 MB NPU	99/144 (68.75%: 30/72 / 69/72)	**144/144 (100.0%**: 72/72 / 72/72)	[97.40%, 100.0%]	**43.02** (**5.68×10^{-14}**)

Table 7 reports the 288-turn live 4-model GGUF evaluation under benchmark serving flags (`--ctx-size 8192 --spec-type ngram-mod --cache-type-k q8_0 --cache-type-v q8_0 --cache-ram 4096`, plus `--cpu-moe --load-mode none --moe-cache-mib 7680` on `26B-A4B`; a fresh `v2` harness run recorded 0.50 s / 0.68 s / 1.53 s / 1.65 s in `logs/test_reports/wo45/web_evidentiary_burden_360_v2_report.json`, and timestamped `nvidia-smi` VRAM captures in `logs/test_reports/wo45/q1_vram_nvidia_smi_evidence.json` measured 2,560 / 4,502 / 10,620 / 11,102 MiB at `ctx=8192`, 2,732 / 4,802 / 10,920 / 11,452 MiB at production `ctx=32768`, and 2,962 / 5,330 /

11,320 / 11,920 MiB at `ctx=65536`). Under the original `WO-44` lexical regex grader (`_HEDGE_OR_CONFLICT_RE`), Arm A (Legacy Static-0.85) scored 99/144 (68.75%, passing $30/72 = 41.7\%$ of Hedge-Required cells) while Arm B (2D Burden-Gated Synthesis) scored 144/144 (100.0%, 36/36 per model). **Methodological limitation and blinded semantic re-grading (`WO-45 R-10 v2/R-11/R-12`, `logs/test_reports/wo45/r10v2_regrade_288_report.json`):** In `WO-44`, Arm B's system prompt explicitly instructed the model to emit the exact legal-standard tokens (*"balance of probabilities"*, *"clear and convincing"*, *"speculation"*, *"dispute/conflict"*) matched by `_HEDGE_OR_CONFLICT_RE`, whereas Arm A received no hedging instruction, no instruction-only control (`Arm C`) was tested, and the 36-item slice excluded items where the NPU hazard prediction differed from the nominal stratum label. When all 288 frozen replies were re-graded blind under a neutral rubric with zero policy-threshold hints by two cross-family subagent judges (`claude-opus-4-6` and `gemini-2.5-flash`, with disagreements adjudicated by `gemini-2.5-pro`; subagent `judge_model`, `temperature`, and `timestamp_utc` self-reported): on 36 disputed-source item pairs (4 models × 9 items), the burden-contract arm disclosed the cross-source conflict in 36/36 (100.0%, 95% Wilson CI $[90.36\%, 100.00\%]$) replies vs. 11/36 (30.56%, CI $[17.99\%, 46.88\%]$) without it (`D4` $\kappa = 0.7990$ unredacted [265/288]; $\kappa = 0.9914$ [287/288] when legal-standard phrases are redacted to `[STANDARD]`, where `D4=YES` is 12/36 [33.33%] vs. 36/36 [100.0%] and composite pass is 12/36 vs. 35/36). On non-disputed items where Lucca's policy requires extra corroboration (36 pairs with 1–2 agreeing sources), blinded judges disagreed on whether hedging was warranted (`D1` $\kappa = 0.2038$ [229/288]: Judge 1 rated unhedged Arm A `OVERCONFIDENT` on 34/36 pairs, whereas Judge 2 and Judge 3 rated 33/36 Arm A replies acceptable and Judge 3 flagged 24/36 Arm B domain-count hedges as `D3=YES` over-hedging), showing that domain-count hedging on agreeing sources is a disputed policy choice rather than an empirically established calibration gain. Across all 144 item pairs, the overall calibration difference ranges from -0.7 pp (120/144 [83.33%] vs. 119/144 [82.64%], McNemar $18 : 19$, $p = 1.00$ when 3-way `D3` splits count as fail) to $+45.8$ pp (77/144 [53.47%] vs. 143/144 [99.31%] under Judge 1 alone), with $+4.9$ pp (136/144 vs. 143/144) under Judge 2 alone and $+14.6$ pp (122/144 [84.72%] vs. 143/144 [99.31%]) under 3-judge majority (neither tie-break rule was pre-specified prior to unblinding, and Judge 3 shared the Gemini family with Judge 2).

Pre-Registered 4-Arm × 5-Model Live v3 Evaluation (`WO-45 G4 v3`, `RM-0..RM-10`, 2,640 Live Turns at `ctx=32768`).

To resolve all eight measurement defects identified in Live v2 (`logs/test_reports/wo45/REVIEW_G4.md`) and unbundle structured conflict context from prompt wording, we executed `WO45-G4-LIVE-V3-PREREG` (`logs/test_reports/wo45/GATE_G4v3.md`, `g4_live_v3_report.json`) across $N = 132$ fresh stratified `WEB360` items (11/cell, `seed=20261112`, 0 item overlap with Live v2; 33 disputed items, 66 nominal burden-met items) ×4 contiguous arms (`A`, `B`, `C`, `D`) × the same 5 local Gemma-4 `Q6_K` checkpoints (2,640 live turns at `ctx=32768`). **Arm D** (`RM-4`) receives the exact same structured evidentiary context block as Arm B (`[Epistemic Burden Contract: ...]`) paired with Arm C's generic non-legal system prompt (no `MUST` disclosure mandate and no legal-standard vocabulary in the system prompt), isolating the detector's conflict signal (`D vs C`) from the mandate and legal phrasing (`B vs D`). Every stage was OpenTimestamps-stamped against public Bitcoin calendar servers and pushed to `origin/main` strictly in causal order (`RM-6(a)` lock `4a6ecdd` → `RM-6(b)` blinded pools `c5ec1fd` → pre-unblinding Judge-3 upgrade amendment `329ed2b` while keys remained sealed off-repo and all `4a6ecdd` locked files remained byte-unmodified → `RM-6(c)` six individual judge commits `e198f1a`–`de5815c` → `RM-6(d)` unblinding and report `4604afd`). Primary grading ran on deterministic arm-identical normalized text (`normalize_g4v3_replies.py`: 0 forbidden lexicon hits across all 2,640 turns; 36.67% 3-way / 57.50% balanced arm-guess accuracy $\leq 60\%$) using three qualification-screened, machine-logged non-Google local `llama-server` families ($T = 0.0$, seed $= 42$, single contiguous process per pool): `Qwen3.8-27B-Ridge-GGUF` (`Alibaba-Qwen`: 66/66 `D4`, 40/40 `D2` on `RM-2`), `phi-4-Q6_K` (`Microsoft-Phi`: 66/66 `D4`, 40/40 `D2`), and `NVIDIA-Nemotron-3.5-Lightning-30B-A3B-IQ4_NL` (`NVIDIA-Nemotron`: 66/66 `D4`, 38/40 `D2`; qualification pairwise `D4` $\kappa = 1.0000$).

Table 8. Pre-Registered 4-Arm × 5-Model Live v3 Results (`WO-45 G4 v3`, $N = 132$ Fresh Items = 2,640 Live Turns at `ctx=32768`, Primary Normalized 2-of-3 Majority, 10,000-Resample Item-Cluster Bootstrap).

Endpoint (Scored Pop.)	Arm A	Arm C (Tip)	Arm D (Det.+C)	Arm B (Full)	D vs. C (p)	B vs. D (p)	Min κ
`D4` Disclosed (165 /arm)	61/165 (37.0%)	85/165 (51.5%)	**164/165** (**99.4%**)	**165/165** (**100.0%**)	+**47.88** pp (0.0001)	+0.61 pp (0.7214)	**0.6983**
`D2` Unfaithful (660 /arm)	6/660 (0.9%)	9/660 (1.4%)	12/660 (1.8%)	9/660 (1.4%)	+0.45 pp (0.6032)	−0.45 pp (0.3958)	0.1210*
`D2` Faithful Sub. (660/arm)	581/660 (88.0%)	612/660 (92.7%)	**648/660** (**98.2%**)	**650/660** (**98.5%**)	+**5.45** pp (0.0001)	+0.30 pp (0.6294)	0.4450*
`D1 ACCURATE` (660 /arm)	532/660 (80.6%)	614/660 (93.0%)	**643/660** (**97.4%**)	628/660 (95.2%)	+**4.39** pp (0.0012)	−**2.27** pp (0.0108)	0.3310*
`D3` Over-Hedge (330/arm)	47/330 (14.2%)	24/330 (7.3%)	18/330 (5.5%)	15/330 (4.5%)	−1.82 pp (0.5108)	−0.91 pp (0.5660)	**0.8811**

Table 8 summarizes the Live v3 findings (*live `D1` and `D2` $\kappa < 0.60$ reflect 93%–99% class-prevalence compression despite 87.2%–97.4% raw pairwise agreement and are **not** claimed judge-robust):

- **Endpoint-Population Inter-Judge Reliability (`RM-9`):** Computed strictly on the 660 disputed turns using reply-only input (`D4` judges never see source snippets), pairwise Cohen's κ on Primary Normalized is **0.8587** (`Qwen3.8 vs Phi-4`, 94.24%), **0.7154** (`Qwen3.8 vs Nemotron-3.5`, 88.94%), and **0.6983** (`Phi-4 vs Nemotron-3.5`, 89.24%; Secondary Raw $\kappa = 0.8627, 0.7189, 0.7152$), clearing $\kappa \geq 0.60$ across every pair. On the 1,320 nominal burden-met turns, `D3` $\kappa = 0.8846, 0.8811, 0.9177$ (98.18%–98.86%).
- **Unbundled Mechanism (`D vs C` vs. `B vs D`):** Supplying the detector's structured conflict context under Arm C's generic prompt (`Arm D`) raises `D4` conflict disclosure by +**47.88** pp ($[+36.36, +60.00]$, $p = 0.0001$, discordant 79 : 0; identical k/N on Secondary Raw), positive and significant ($p = 0.0001$) under every single judge individually (`Qwen3.8`: +52.73 pp [87 : 0], `Phi-4`: +42.42 pp [70 : 0], `Nemotron-3.5`: +36.97 pp [61 : 0]) and every individual Gemma-4 checkpoint (`E2B-heretic` +69.70 pp, `E4B-heretic` +42.42 pp, `E4B-stock` +45.45 pp, `12B-heretic` +48.48 pp, `26B-A4B-heretic` +33.33 pp, all exact McNemar $p \leq 9.77 \times 10^{-4}$). By contrast, adding Arm B's `MUST` mandate and legal-standard vocabulary on top of Arm D's contract context (`B vs D`) yields a null +0.61 pp ($[0.00, +1.82]$, $p = 0.7214$, 1 : 0) on `D4` (98.8% of the +48.48 pp bundled `B vs C` gain [80 : 0, $p = 0.0001$] comes from `Arm D` alone) while reducing `D1 ACCURATE` calibration by −**2.27** pp ($[-4.09, -0.61]$, $p = 0.0108$; 90.30% vs. 95.76% on the 33 non-disputed below-nominal-burden items), leaving `B vs C` non-significant on `D1` (+2.12 pp, $p = 0.1556$). Both `Arm D` and `Arm B` eliminate `NO_ANSWER` blank refusals on `Tier 4` web snippets (0/660 vs. 73/660 [11.06%] Arm A and 39/660 [5.91%] Arm C), raising substantive faithful answering by +5.45 pp (`D vs C`, $p = 0.0001$) and +5.76 pp (`B vs C`, $p = 0.0001$). Matched `E4B-stock` vs. `E4B-heretic` (`R-17`) score 33/33 vs. 33/33 ($p = 1.0$) on `D4` in both `Arm D` and `Arm B`.

Archived Historical 3-Arm × 5-Model Live v2 Evaluation (`WO-45 G4`, `R-11..R-17`, 1,980 Live Turns at `ctx=32768`; Superseded by `G4 v3` Above).

Known measurement defects in G4 v2 (fixed in v3 — `logs/test_reports/wo45/REVIEW_G4.md`):

- **F-1:** Cited all-turn `D4` $\kappa = 0.77$ (1,980 turns) includes 1,485 non-disputed `N/A`s; on the 495 disputed turns `D4` is scored on, `J1--J2` binary $\kappa =$ **0.064** (raw 327/495).
- **F-2:** `Mistral-Nemo` marks `D4=YES` on 486/495 disputed turns (`A`: 157, `B`: 165, `C`: 164) and `D2=CORRECT` on 1,965/1,980, grading snippet conflict rather than the reply.
- **F-3:** Arm B remains identifiable in 388/660 replies after redaction (`[STANDARD]` occurs 496 times only in Arm B; `threshold`, `standard of proof`, and `burden` are unredacted; 7/165 disputed B replies are marker-free).
- **F-4:** Judge 1 self-reports `gemini-2.5-pro` (`claude-3-7-sonnet` on one shard vs. `flash` in the lock), is not machine-logged, and shares the Google vendor family with all 5 Gemma generators while driving the effect.
- **F-5:** `B vs C` bundles three changes: contract context, a `MUST` disclosure instruction, and legal vocabulary (unbundled via `Arm D` in v3).
- **F-6:** Only `live_v2_prereg_lock.json` had an external OpenTimestamps proof; pools, judge outputs, and analysis were first pushed together at `00:31Z`.
- **F-7:** The `R-41` 3/15 figure comes from 3 post-hoc selected items with a turn-level CI; the underlying NPU over-escalation (`R-40 v2`: 73/73 true over-escalations) is retrained in `R-42` (`RM-7`) before v3.
- **F-8:** Judge 3 was sharded (8×208) and shard 7 re-run without either procedure pre-specified in the lock.

To test deterministic burden-gate conflict detection against generic prompt instruction (`G4-3`), eliminate classifier-agreement item selection bias (`R-12`), include a matched stock non-abliterated checkpoint (`R-17`), and enforce cryptographic pre-registration (`R-14`, `R-23`), we locked `logs/test_reports/wo45/live_v2_prereg_lock.json` (`ee379f91...`, anchored via OpenTimestamps in `live_v2_prereg_lock.json.ots` and pushed in commit `554c7c6` prior to generation), hashing all evaluation code including `scripts/wo45/analyze_wo45_live_v2.py`, neutral judge prompts (`0` banned policy-term hits), and $N = 132$ unconditioned stratified items (11/cell, `seed=20261111`, 33 disputed, 31 pre-declared from the `R-40 v2` 73-item escalation set). Five local Gemma-4 `Q6_K` models (`E2B-heretic` [2,732 MiB, 0.40 s], `E4B-heretic` [4,802 MiB, 0.56 s], stock `gemma-4-E4B-it-Q6_K.gguf` [4,970 MiB, 0.56 s], `12B-heretic` [10,920 MiB, 1.24 s], and `26B-A4B-heretic` [11,452 MiB, 1.27 s] at production `ctx=32768`) each generated 396 contiguous-arm turns across Arm A (plain baseline), Arm B (2D burden-contract gate), and Arm C (Arm A context plus a generic non-legal instruction: *"If the sources are few, low-quality, or disagree, say so and express appropriate uncertainty"*). Blinded unredacted and `[STANDARD]`-replaced pools (`689c381`), raw judge outputs from Judge 1 Google Gemini (`Model="flash"` subagents; self-reported metadata, Defect F-4), Judge 2 Mistral AI (machine-logged local `Mistral-Nemo-Instruct-2407-Q5_K_M`, $T = 0.0$, seed $= 42$; non-discriminating, Defect F-2), and Judge 3 Adjudicator Alibaba Qwen (machine-logged local `Qwen3.8-27B-Ridge-GGUF`, $T = 0.0$, seed $= 42$ breaking `J1 != J2` ties; sharded 8×208, Defect F-8; `377ed7a`), and unblinding (`b9c7efb`, `logs/test_reports/wo45/g4_live_v2_report.json`; pushed together at `00:31Z`, Defect F-6) yielded the following historical v2 counts (preserved verbatim alongside the `F-1..F-8` disclosures, superseded by `G4 v3` above):

- **Co-Primary 1 — `D4` Conflict Disclosure on Disputed Items ($N_{\text{eff}} = 33$ items, 165 turns/arm; disputed-population `J1 vs J2` binary $\kappa =$ 0.064 [327/495], Defect F-1):** Under 3-family majority (10,000-resample item-cluster bootstrap), Arm B discloses cross-source conflicts on 163/165 (98.79%, CI $[95.69\%, 99.67\%]$) turns vs. generic-instruction Arm C on 121/165 (73.33%, CI $[66.11\%, 79.49\%]$, $\Delta(B - C) = +25.45$ pp $[+15.76, +35.76]$, $p = 0.0001$, discordant 43 : 1; bundles contract context, `MUST`, and legal vocabulary, Defect F-5) and plain Arm A on 104/165 (63.03%, CI $[55.45\%, 70.02\%]$, $\Delta(B - A) = +35.76$ pp $[+25.45, +46.67]$, $p = 0.0001$, 61 : 2; Arm C vs. Arm A: $+10.30$ pp $[-0.61, +20.61]$, $p = 0.0668$, 32 : 15; per-model counts: `E2B-heretic` 31/33 vs. 24/33; `E4B-heretic` 33/33 vs. 24/33; `E4B-stock` 33/33 vs. 24/33; `12B-heretic` 33/33 vs. 21/33; `26B-A4B-heretic` 33/33 vs. 28/33). Under `[STANDARD]` replacement (`G4-4`; Arm B remained identifiable in 388/660 replies, Defect F-3), counts are 163/165 (98.79%) vs. 129/165 (78.18%, $+20.61$ pp $[+12.12, +30.30]$, $p = 0.0001$, 35 : 1). Single-judge sensitivity on `D4 B vs C` spans $+36.36$ pp under Judge 1 Gemini alone (157/165 vs. 97/165, $p = 0.0001$)

to +0.61 pp under Judge 2 Mistral-Nemo alone (165/165 vs. 164/165, $p = 0.7330$, marking 486/495 disputed turns `YES`).

- **Co-Primary 2 — `D2` Factual Correctness ($N_{\text{eff}} = 132$ items, 660 turns/arm; `J1 vs J2` raw agreement 94.24%, binary $\kappa = 0.1845$; Mistral marked 1,965/1,980 `CORRECT`, Defect F-2):** Arm B scores 656/660 (99.39%, CI [98.45%, 99.76%]) vs. Arm C 629/660 (95.30%, +4.09 pp [+1.67, +6.82], $p = 0.0004$, 31 : 4 ; +4.85 pp Gemini alone, 0.00 pp Mistral alone) and vs. Arm A 593/660 (89.85%, +9.55 pp [+6.21, +13.18], $p = 0.0001$, 66 : 3).
- **Secondary & Negative Results (`D1`, `R-16`, `R-17`):** Secondary `D1` epistemic signalling remains low-agreement across judges (`J1 vs J2` binary $\kappa = 0.0686$) and shows no Arm B advantage over Arm C (612/660 [92.73%] vs. 627/660 [95.00%], $\Delta = -2.27$ pp [−5.15, +0.76], $p = 0.1444$; on the 33 non-disputed below-nominal-burden items, Arm B scores 124/165 [75.15%] vs. 139/165 [84.24%] Arm A and 144/165 [87.27%] Arm C as judges penalize domain-count hedging on non-conflicting sources as `UNDERCONFIDENT`). Matched `E4B-stock` vs. `E4B-heretic` (`R-17`) are identical on Arm B (33/33 `D4`, 132/132 `D2`, McNemar $p = 1.0$) and Arm C `D4` (24/33 both, $p = 1.0$), while `E4B-stock` exhibits higher spontaneous refusal/over-hedging on unguarded Arm A (`D3`: 18/66 [27.27%] vs. 6/66 [9.09%], McNemar 12 : 0, $p = 0.00049$).

5. Compounded System Integration & Modular Decomposition (`WO-42`–`WO-43`)

Upgrading the three NPU classifiers, adding 3-tier anti-bot web extraction (`scrapling v0.4.6` with `curl_cffi` TLS fingerprint impersonation and `sanitize_html_against_prompt_injection` CSS/comment stripping in `src/lucca/sandbox/web_tool.py`), and decomposing the orchestrator and domain god-objects in Work Order 43 (`v1.51.0`: `orchestrator.py` [2,195 LOC], `turn_pipeline.py` [1,559 LOC], `speech_stream_pipeline.py` [946 LOC], `tool_dispatcher.py` [975 LOC], `deduction_runner.py` [1,378 LOC], `code_preflight_service.py` [366 LOC], `memory_commit_service.py` [844 LOC], `cascade.py` [775 LOC], `cascade_intent.py` [781 LOC], `cascade_substrates.py` [1,008 LOC], `output_supervisor.py` [475 LOC], `supervisor_lexicon.py` [657 LOC], `action_claim_grammar.py` [607 LOC], `receipt_truth_matcher.py` [488 LOC], `cegis_unshackled.py` [575 LOC], `cegis_grid_diff.py` [389 LOC], `cegis_sandbox.py` [1,102 LOC], `cegis_mutator.py` [211 LOC], and `walton_schemes/`) resolved four downstream bottlenecks in the cognitive architecture (`tests/unit/test_wo42_compounded_enhancements.py`):

1. **Multi-Category Compound Tool Shortlisting (`src/lucca/core/tool_shortlister.py:225--297`):** Prior to `WO-42`, `shortlist_tools()` discarded all secondary task categories whenever more than one category was activated, starving two-sentence prompts (e.g., *"Search the web for CVE-2026-4096 and write a summary to audit.md"*) of workspace tools. Packing secondary activated categories into the remaining slots under the hard ≤ 12 tool cap preserves all 6 `cognitive_core` tools plus both task categories.
2. **Deep-Extract Search & Citation Provenance (`src/lucca/sandbox/service.py:616`, `src/lucca/core/tool_dispatcher.py`):** Exposing `deep_extract: bool` on `search_web` routes multi-source queries to `research_topic` without exceeding the 24-tool registry cap (`MAX_REGISTRY_TOOLS = 24`), while registering scraped `"sources"` URLs with `OutputSupervisor.register_allowed_urls` so cited URLs are not flagged as fabricated citations.
3. **Sentence-Filtered Cross-Reference Corroboration (`src/lucca/core/deduction_runner.py`, `src/lucca/core/corroboration.py:666`):** When `NeuralHazardClassifier` triggers `CROSS_REFERENCE_CMD`, `CorroborationEngine.compute_proposition_alignment` checks whether candidate evidence contains negation or adversative markers (`_RE_NEGATION_POLARITY`, `_RE_ADVERSATIVE_DEFEATER`). Passing an entire 3,500-character scraped article body caused false conflicts whenever unrelated paragraphs contained words such as *"not"* or *"however"*, whereas passing 150-

character search snippets truncated version numbers and dates. Extracting only the top 1–2 sentences with highest claim-token overlap per independent domain (`_extract_claim_aligned_sentences`) preserves per-domain receipt independence while preventing unrelated paragraph negations from defeating corroboration.

4. **NLI Entailment Wiring in Offline Skill Induction (`src/lucca/reasoning/induction.py:402--465`, `reflection.py:373`):** Passing `verifier: SemanticVerifierPort` from `DreamRetrospective` into `RuleInducer.mine_patterns()` activates bidirectional NLI entailment clustering in `cluster_phrasings()` for paraphrased multi-sentence tool arguments whose token-set Jaccard similarity falls below 0.60.

6. Limitations and Residual Error Analysis

1. **Residual False Negatives ($6/471 = 1.27\%$ on $N = 2{,}022$):** Inspection of the 6 missed positive claims in `logs/test_reports/wo42_large_sample_2020_report.json` shows that all 6 come from the 1,000-turn sampled corpus (`W751..W1000`) where the model implied completion through an indirect state consequence without an explicit action verb or status token (e.g., *"Your inbox is now completely empty and ready for Monday"* in response to an email archive prompt).
2. **Residual False Positives ($9/1{,}551 = 0.58\%$ on $N = 2{,}022$):** The 9 false alarms occur on hypothetical troubleshooting walkthroughs where the assistant writes first-person illustrative steps (*"First, I updated the registry key in my test environment..."*) without fencing them in code blocks.
3. **Linear Latency Scaling with Clause Count:** Because `NeuralActionClaimVerifier` evaluates the full response plus each extracted clause sequentially under a thread lock on the Intel NPU, mean verification latency scales from 1.31 ms on 1-sentence outputs to 2.95 ms (P95 3.67 ms) on 3–4 sentence outputs. On 10+ sentence long-form essays, clause-by-clause NPU scoring can reach 8–12 ms; batching clause tensors into a single static-shape NPU call is left for future work.
4. **Single-Workstation Measurement Scope:** All latency, VRAM, and live 4-model measurements were recorded on a single Windows 11 workstation (Intel Core Ultra 9 275HX, Intel AI Boost NPU, NVIDIA RTX 5080 Laptop GPU 16 GB GDDR7) and are labeled *Author-measured* until independently reproduced on external hardware.

7. Conclusion

Lexical speech-regex supervisors cannot generalize to open-vocabulary verb lemmas (29.5% unseen-verb recall in Paper 02 [2]), while whole-paragraph neural classifiers degrade when a single fabricated action claim or task request is embedded inside multi-sentence prose (75.0% on `Blind-300`). By combining syntactic clause decomposition with deterministic receipt set-membership verification, max-hazard lattice pooling, and task-priority intent pooling, three 82.1M-parameter `nli-MiniLM2-L6-H768` classifiers running in FP16 on an Intel AI Boost NPU (0 MB dGPU VRAM, 1.22–2.95 ms) achieve 99.26% accuracy (98.73% recall, 0.58% FPR) across $N = 2{,}022$ action-claim turns, 96.67% single-pass / 100.0% post-fix accuracy on the pre-registered `WO42_LOCKED_HOLDOUT_150` benchmark, and 100/100 functional coherence across all four local Gemma-4 model scales.

References

1. Sean Maxwell. Cryptographic Provenance Gates Mitigate Multi-Session Memory Poisoning in Autonomous Agents. *Zenodo Preprint*, DOI: 10.5281/zenodo.23901452, October 2026.
2. Sean Maxwell. Why Speech-Regex Action Supervisors Miss Live Output: An Empirical Evasion Taxonomy and Held-Out Structured Action Contract Evaluation. *Zenodo Preprint*, DOI: 10.5281/zenodo.23901605, October 2026.
3. Sean Maxwell. Closing the Epistemic Loop Across the Five Layers of Agent Memory: Cryptographic Provenance, Paraconsistent Bilattices, and Z3-Quarantined Procedural Induction. *Zenodo Preprint*, DOI: 10.5281/zenodo.23901683, November 2026.

4. Pengcheng He, Jianfeng Gao, and Weizhu Chen. DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv: 2111.09543.
5. Dmitry Kobak, Rita González-Márquez, Emőke-Ágnes Horvát, and Jan Lause. Delving into LLM-assisted writing in biomedical publications through excess vocabulary. *Science Advances*, 2025. arXiv: 2406.07016.
6. Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947.